

Exercises, sheet 5.

1. Solve Exercise 8.1.
2. Prove Lemma 8.10.
3. Solve Exercise 8.16.
4. Solve Exercise 8.22. In particular, if $M \models T$, then we denote by $M^{\text{eq}} \models T^{\text{eq}}$ its unique (up to a unique isomorphism) lift to a model of T^{eq} .
5. Show that if $M \models T$ is κ -saturated and $N = (\varphi/E)(M)$ is an L' -structure interpreted in M , then N is also κ -saturated.
6. Let T' be an L' -theory, with $L' = L \cup \{P\}$ for some n -ary predicate P . Assume that if $A, B \models T'$ are such that $A|_L = B|_L$, then for all $c \in A^n$ we have

$$A \models P(c) \iff B \models P(c).$$

Show that there is an L -formula $\varphi(x_1, \dots, x_n)$ such that

$$T' \vdash P(x_1, \dots, x_n) \leftrightarrow \varphi(x_1, \dots, x_n).$$

This result is called Beth definability theorem.

7. Let $\mathfrak{C}$ be a κ -monster model of a complete theory T and let $\varphi(x, a)$ be an L -formula. Show that there is an element $a' \in \mathfrak{C}^{\text{eq}}$ such that for all $\sigma \in \text{Aut}(\mathfrak{C})$ we have

$$\sigma(\varphi(\mathfrak{C}, a)) = \varphi(\mathfrak{C}, a) \iff \sigma(a') = a'.$$

Show that there is an L^{eq} -formula $\psi(x, y)$ such that $\varphi(\mathfrak{C}, a) = \psi(\mathfrak{C}, a')$. If we write $X = \varphi(\mathfrak{C}, a) \subseteq \mathfrak{C}^x$, an element a' with the considered property is called the *canonical base* of X .

8. Show that if a', a'' are canonical bases of a definable set $X \subseteq \mathfrak{C}^x$, then there is a definable function f whose domain contains a' such that $f(a') = a''$.